Jesus Garcia

garci1294@gmail.com · +1 (612) 723-5319 · Minneapolis, MN

EXPERIENCE

Mayo Clinic **Rochester, MN**
Senior AI/HPC Systems Engineer *Nov 2025 – Present*

- Operate three NVIDIA DGX SuperPOD clusters totaling 512 GPUs across 64 nodes, plus a 208 GPU general purpose HPC system, on a five person team responsible for provisioning, scheduling, and daily support of Mayo Clinic research computing.
- Drove GPU utilization from under 50 percent to a sustained 80–85 percent across clusters by building the reporting behind it, mining sacct and sreport for per lab usage patterns, standing up an agent to track them continuously, and putting the results in front of labs so they right sized their own requests.
- Helped plan a new Kubernetes and Run:ai cluster for GPU scheduling, adding project quotas, fair share, and fractional GPU sharing alongside the Slurm based DGX systems.
- Built a bubblewrap based sandbox that brought agentic AI workloads onto protected clinical infrastructure, wiring a managed LLM gateway key into supported agents for one step setup and giving labs the same primitive to wrap agents they write themselves. Now open to most users across all clusters.
- Administer Lustre on Dell hardware for the B300 cluster and DDN storage for B200 systems, running NCCL collective and Hugging Face model benchmarks to validate sustained throughput before releasing capacity to production research.
- Planned and helped deliver stateless provisioning with Warewulf on the general purpose HPC system, making node rebuilds repeatable and clearing the path for the H200 and high memory expansion.
- Contributed to the buildout and expansion of Mayo B200 and B300 DGX clusters alongside Mark III Systems and NVIDIA field engineers, covering network planning, configuration, and the operating documentation the team runs against today.
- Support 50+ research labs running reproducible multi node GPU pipelines across AlphaFold protein structure prediction, computational pathology, and memory intensive genomics.

University of Colorado Boulder, Research Computing **Boulder, CO**
High Performance Computing Engineer *Apr 2024 – Nov 2025*

- Led the Alpine cluster expansion, adding 64 NVIDIA GPUs across 16 nodes, 8 with H200 and 8 with RTX Pro 6000, alongside 17 CPU nodes carrying 2,176 cores, sized specifically to cut AI and ML job wait times for the campus research community.
- Owned the full path to that expansion, from datacenter facility planning and Ceph storage architecture through Dell vendor negotiation and budget modeling, and presented the proposal to institutional leadership to secure approval.
- Led Puppet and Ansible configuration management across several hundred Linux nodes on the Alpine and Blanca clusters, a shared environment jointly funded by three universities and the National Science Foundation, delivering repeatable builds and same day scale up for genomics, climate modeling, and imaging research.
- Built Git driven pipelines that trigger Ansible playbooks with no cluster downtime, giving investigators fresh software stacks on demand rather than waiting for a maintenance window.
- Onboarded NVIDIA L40S and H100 accelerators into production, tuning Slurm partition layout, QoS, and CUDA configuration so new GPU capacity reached researchers quickly, and validated every new GPU node against TensorFlow and PyTorch benchmark suites to confirm driver and CUDA library compatibility and pull faulty cards early.
- Designed container workflows on Docker, Podman, and a Kubernetes lab cluster, replacing hand built environment modules with versioned TensorFlow and PyTorch images so researchers pulled a working GPU stack in minutes instead of waiting on a manual rebuild.

University of Minnesota, Minnesota Supercomputing Institute **Minneapolis, MN**
Linux Systems DevOps Engineer *Mar 2022 – May 2024*

- One of four engineers who built the LORIS neuroimaging data platform for the Masonic Institute for the Developing Brain, serving 50+ research institutions across the US and Canada.
- Engineered petabyte scale storage and transfer supporting the Center for Mesoscale Connectomics, moving millions of gigabytes of imaging data between institutions worldwide on MSI infrastructure.
- Supported the first data release of the NIH Brain Development Cohorts platform, integrating the ABCD and HBCD studies into a single access controlled environment.
- Automated production deployments with Puppet modules under Git version control, shortening release cycles and raising reliability across MSI research systems.
- Secured HIPAA protected MRI and genomic data with role based access control, and integrated AWS S3 for archival with lifecycle policies that cut storage cost while improving durability.

University of Minnesota, Office of Information Technology **Minneapolis, MN**
Software Packaging Engineer *Aug 2018 – Mar 2022*

- Managed the lifecycle of Windows and macOS virtual machines on VMware vSphere and MacStadium Orka, producing golden images and automated snapshots that kept lab builds current and secure.

- Wrote AutoPkg recipes and Munki manifests in Git, enabling nightly automated package builds and peer review, and maintained runbooks that cut new engineer ramp up from four weeks to one.

PROJECTS

Plain Theory Labs · Founder **plaintheory.org · github.com/plain-theory-labs**

- Built and released an open framework that produces a single deterministic efficiency score for AI compute clusters, combining seven engines across GPU utilization, thermal efficiency, carbon accounting, scheduler policy, and hardware to workload fit, with published weights so results reproduce independently.
- Wired the framework to ingest live cluster telemetry from DCGM, Slurm, Kubernetes, and inference servers, so assessments run on collected data rather than self reported numbers.
- Validated the scoring model against public cluster traces from MIT Supercloud, Alibaba Helios, and Microsoft Philly, publishing engines and methodology under MIT and CC BY 4.0.

University of Minnesota Brain Tumor Program · Lead Computational Biologist **Sep 2019 – Present**

- Run GPU accelerated single cell genomics at scale, training scVI and UniTVelo models in PyTorch and CUDA on Slurm clusters to trace developmental lineages across glioma and chordoma cohorts.
- Manage FASTQ data from BGI and 10x Genomics through Cell Ranger, STARsolo, and Kallisto pipelines on Slurm, producing count matrices spanning 100+ bulk RNA sequencing patients and 21 single nuclei samples.
- Harmonize epigenomics and whole genome sequencing datasets with collaborators at several universities, versioning pipelines in Nextflow and Git so analyses rerun reproducibly.

TECHNICAL SKILLS

Accelerated Computing NVIDIA DGX SuperPOD (H100, H200, B200, B300), RTX Pro 6000, L40S, A100, CUDA Toolkit, NCCL, DCGM, GPU benchmarking and node validation

Schedulers & Orchestration Slurm, Run:ai, Kubernetes, Docker, Podman, NVIDIA Base Command Manager (cmsh), Open OnDemand

Parallel & Scale Out Storage Lustre, Ceph, DDN, Isilon, Globus data transfer

Provisioning & Automation Warewulf stateless provisioning, Puppet, Ansible, Git, GitHub, Nextflow

Observability & Telemetry Prometheus, Grafana, DCGM exporter, Splunk, Elastic, Nagios

Programming & Scripting Python (PyTorch, TensorFlow), Bash, CUDA C, C and C++ (reading, patching, and building HPC software from source), Perl, R (Bioconductor)

Scientific Computing Workloads LORIS, AlphaFold, Cell Ranger, STARsolo, Kallisto, scVI

Platforms & Cloud RHEL, Rocky Linux, CentOS, Windows Server, AWS (EC2, EKS, S3), Microsoft Azure, Google Cloud Platform

Security & Isolation bubblewrap sandboxing, HIPAA controls, role based access control

EDUCATION

University of Minnesota **Minneapolis, MN**
Master of Science 2026
Bachelor of Science 2018

PUBLICATIONS

Garcia Garcia J, Grillo S, Cao Q, Brunstein CG, Arora M, MacMillan ML, Wagner JE, Weisdorf DJ, Holtan SG. *Low 5-Year Health Care Burden After Umbilical Cord Blood Transplantation.* Blood Advances, 2021; 5(3): 853–860. doi:10.1182/bloodadvances.2020003369

Garcia Garcia J, Grillo S, Cao Q, Brunstein CG, Arora M, MacMillan ML, Wagner JE, Weisdorf DJ, Holtan SG. *5-Year Health Care Burden After Allogeneic Hematopoietic Stem Cell Transplantation (HSCT): Impact of Graft Source.* Blood, 2018; 132 (Supplement 1): 367. doi:10.1182/blood-2018-99-11320

CERTIFICATIONS, MEMBERSHIPS & AWARDS

Certifications NVIDIA DLI Certified Instructor · Generative AI with Diffusion Models · NASA Transform to Open Science (TOPS) · NASA L'SPACE Program · AI Unlocked (NAIRR)

Memberships National Artificial Intelligence Research Resource (NAIRR) Pilot · Society of Hispanic Professional Engineers (SHPE) · American Association for Cancer Research (AACR) · American Society for Cell Biology (ASCB)

Awards Outstanding Abstract Achievement Award, American Society of Hematology (2018) · President's Student Leadership and Service Award, University of Minnesota (2018)

Languages English (bilingual) · Spanish (native)